



UNIVERSITÀ
DI TORINO

Electrical signals in neurons: local and action potential

MATTEO DIANO, PH.D.

COMPUTATIONAL APPLICATION TO COGNITIVE NEUROSCIENCE

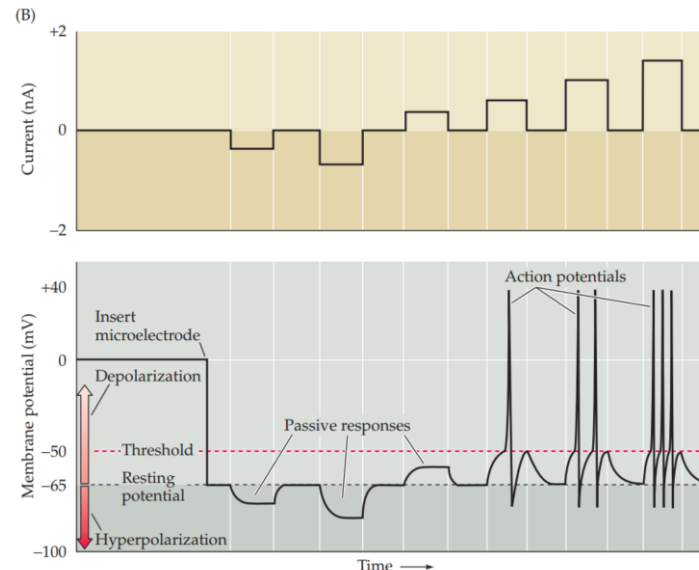
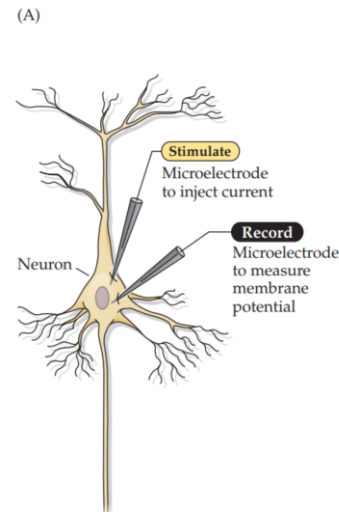
REMINDER

Hyperpolarization

- The current produced makes the membrane potential more negative.
- The membrane potential changes in proportion to the amount of current applied.

Depolarization

- Current of opposite polarity is applied.
- The membrane potential becomes more positive than the resting potential (and may exceed the threshold potential).



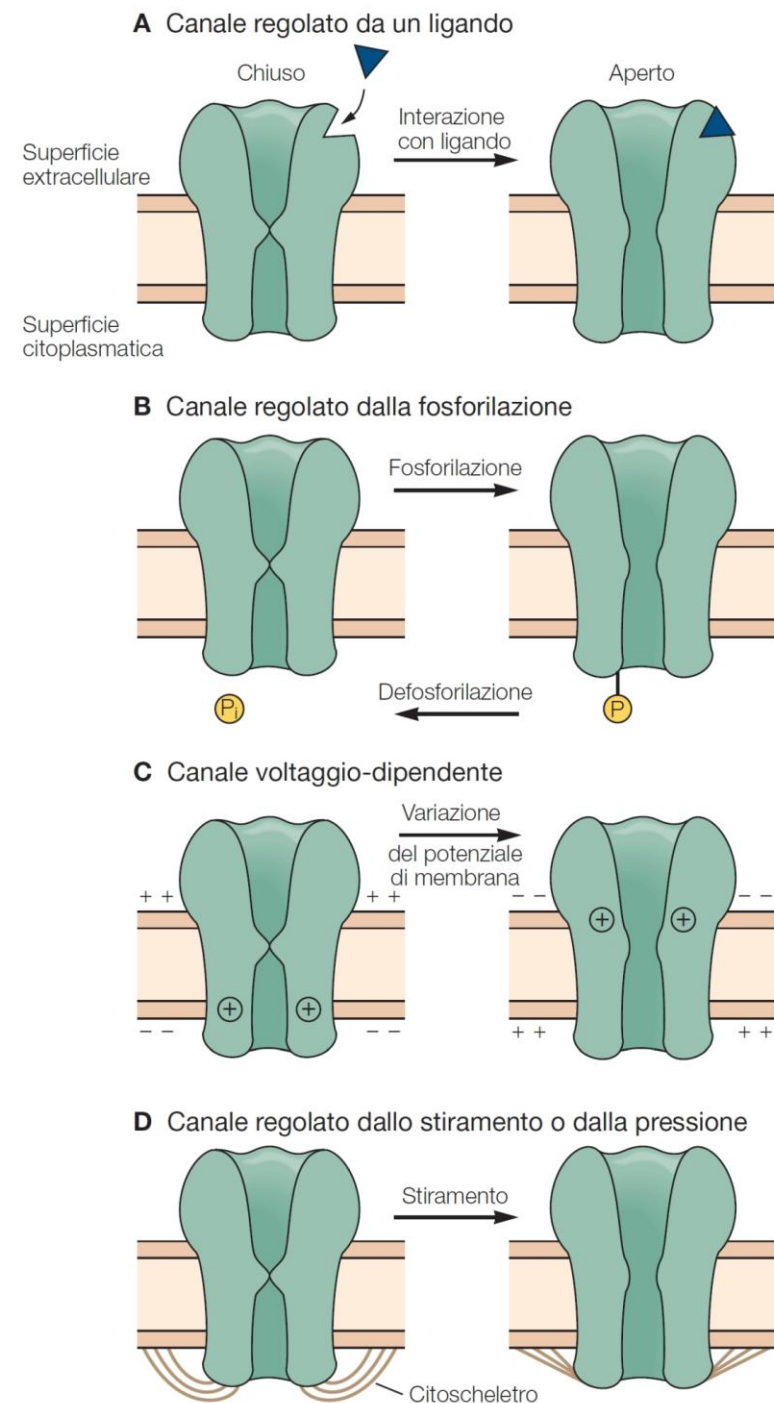
Before starting: ion - channel

Permanently open channels (resting potential):

- These channels are always open.
- They allow continuous ion flow that helps set the baseline electrical conditions of the membrane.

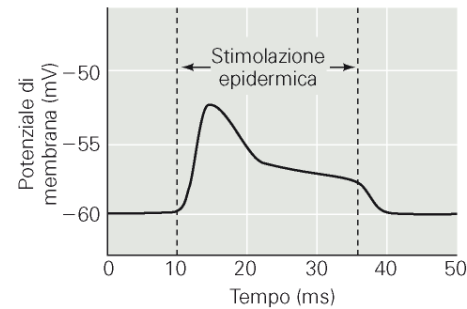
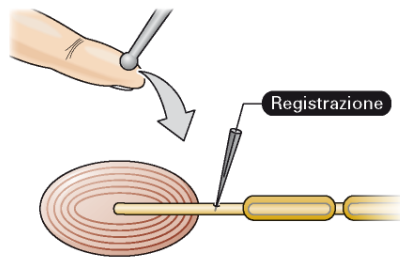
Gated channels (can open or close depending on conditions):

- **Ligand-dependent:** open or close when a specific chemical substance binds.
- **Phosphorylation-dependent:** their state is controlled by the addition or removal of a phosphate group.
- **Voltage-dependent:** open or close depending on changes in the membrane potential.
- **Membrane stretch-dependent:** respond to mechanical deformation of the membrane.

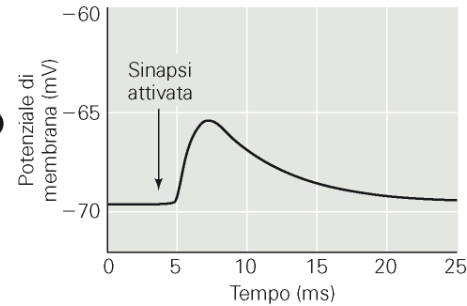
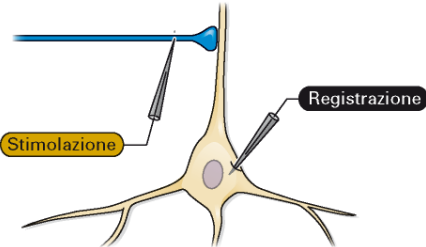


Characteristics of local and action potentials

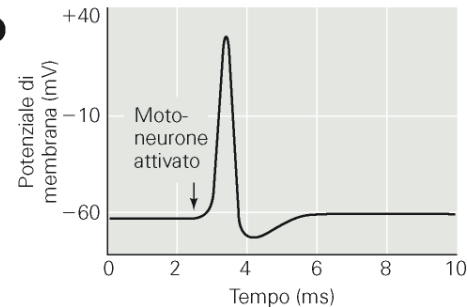
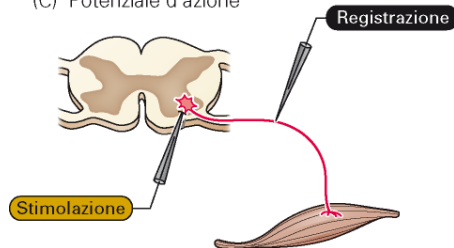
(A) Potenziale di recettore



(B) Potenziale sinaptico



(C) Potenziale d'azione



A) and B) Receptor and synaptic potentials (= types of local potentials):

- Local signals
- Small amplitude
- Short duration
- Gradual summation
- Passive propagation
- Depolarizing or hyperpolarizing effect

C) Action potential:

- Propagated signals
- Large amplitude
- Short duration
- All-or-none
- Active propagation
- Only depolarizing effect

Characteristics of local and action potentials

Tabella 2-1 Confronto fra le caratteristiche dei segnali locali (passivi) e propagati

Tipo di segnale	Ampiezza (mv)	Durata	Sommazione	Effetti del segnale	Tipo di propagazione
Segnali locali (passivi)					
Potenziali di recettore	Piccola (0,1-10)	Breve (5-100 ms)	Graduale	Iperpolarizzanti o depolarizzanti	Passiva
Potenziali sinaptici	Piccola (0,1-10)	Da breve a lunga (5 ms-20 min)	Graduale	Iperpolarizzanti o depolarizzanti	Passiva
Segnali propagati (attivi)					
Potenziali d'azione	Grande (70-110)	Breve (1-10 ms)	Tutto-o-nulla	Depolarizzanti	Attiva

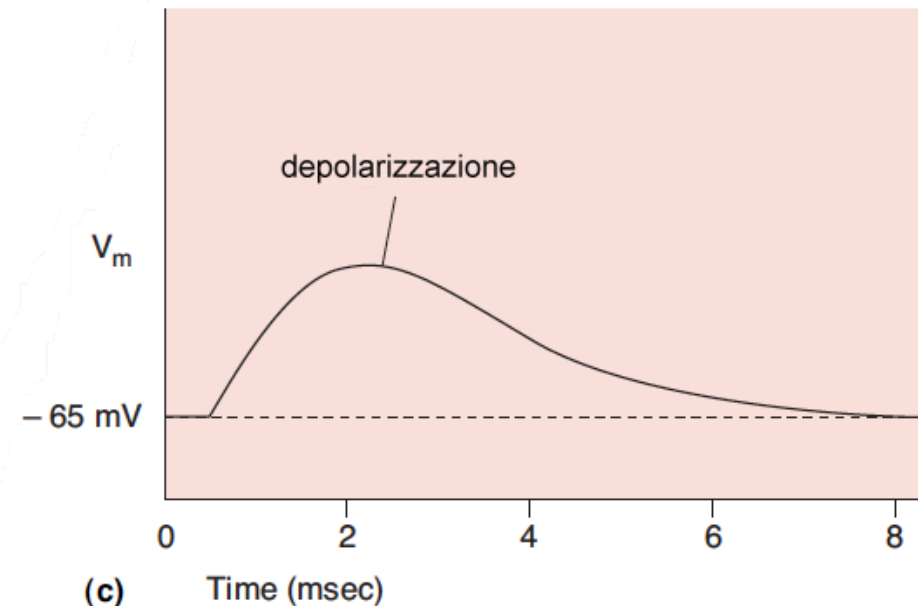
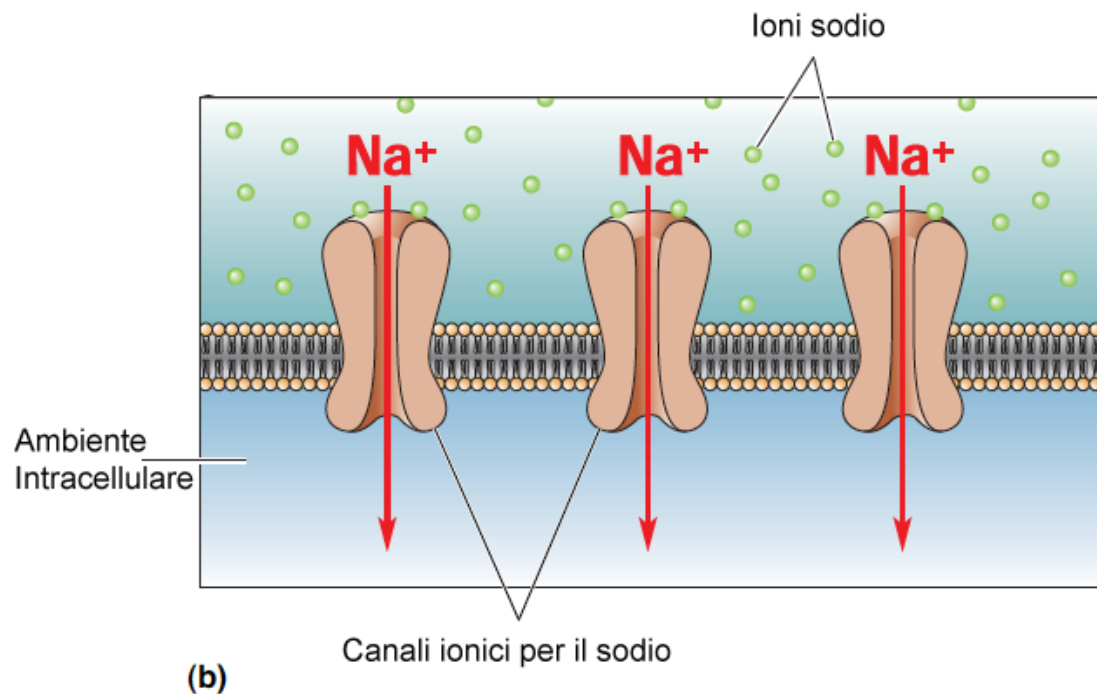
Characteristics of local potentials – some mechanisms

Depolarization: the membrane potential becomes more positive

(e.g., -10 , 0 mV)

influx of positive ions (Na^+ , Ca^{2+})

efflux of negative ions (Cl^-)



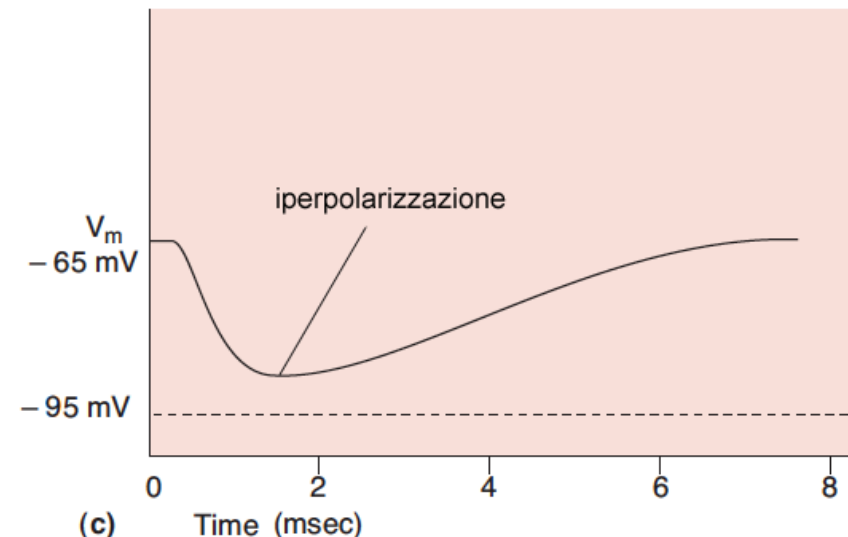
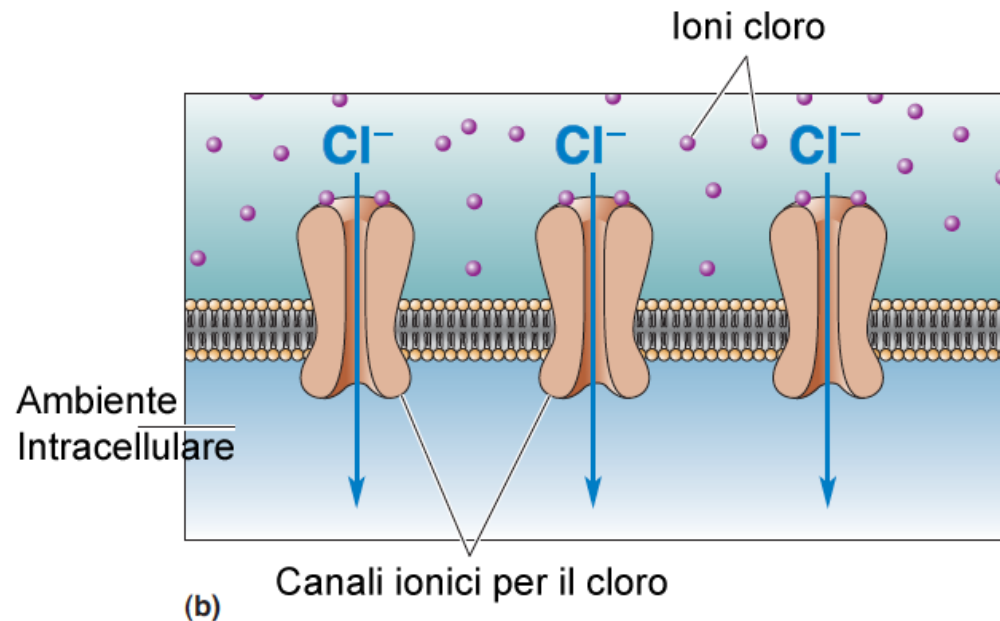
Characteristics of local potentials – some mechanisms

Hyperpolarization: the membrane potential becomes more negative

(e.g., -80 , -90 mV)

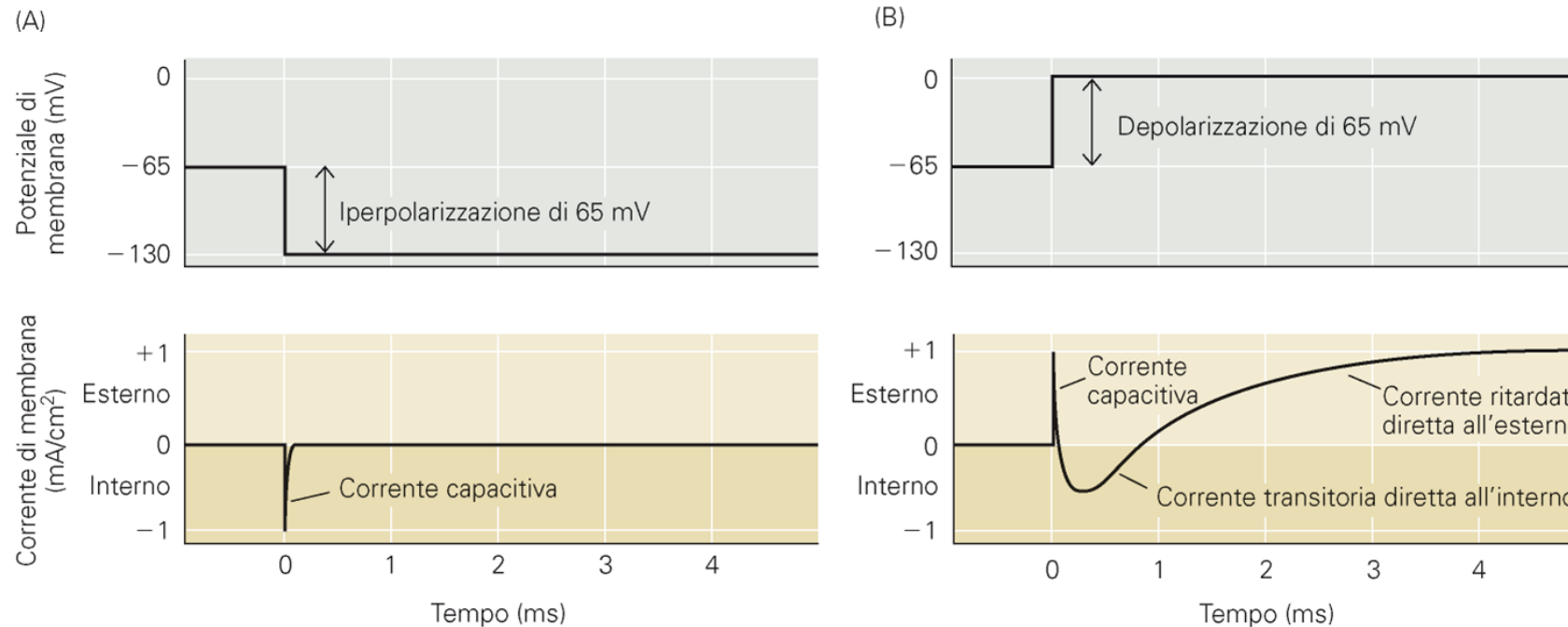
efflux of positive ions (K^+)

influx of negative ions (Cl^-)



Two types of voltage-dependent ionic currents

The voltage clamp technique allows the observation of changes in membrane currents while the membrane potential is held under control, enabling the separation of capacitive and ionic components and the characterization of voltage-dependent conductances

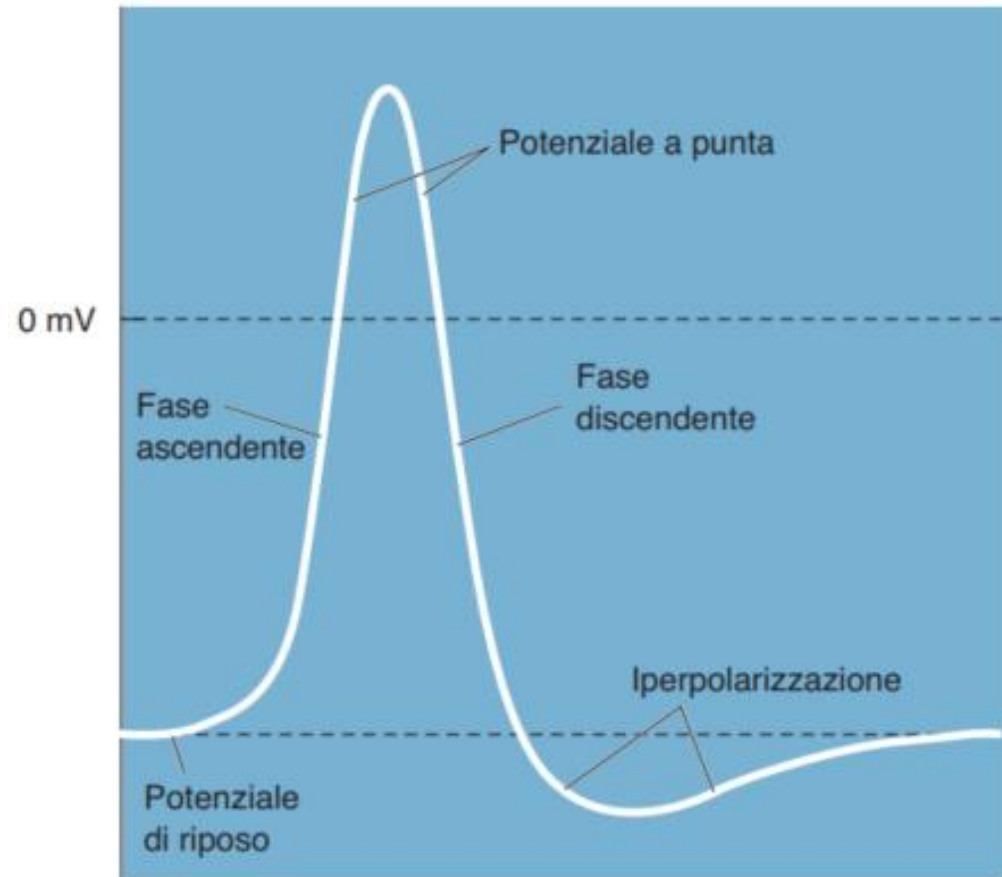


1. The early current (transient inward) is determined by the influx of Na^+ into the neuron.
2. The delayed current (sustained outward) is determined by the efflux of K^+ from the neuron.

The Action Potential

Everything starts from the resting potential:

- Ascending phase
- Peak potential (overshoot)
- Descending phase
- Hyperpolarization (undershoot)



The Action Potential (live)



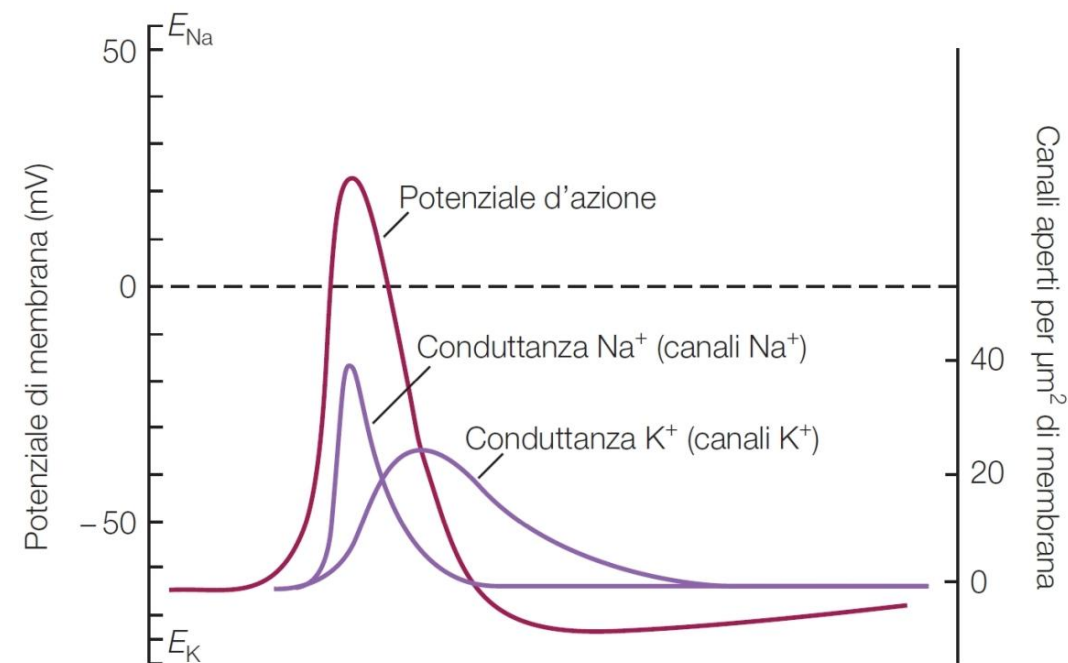
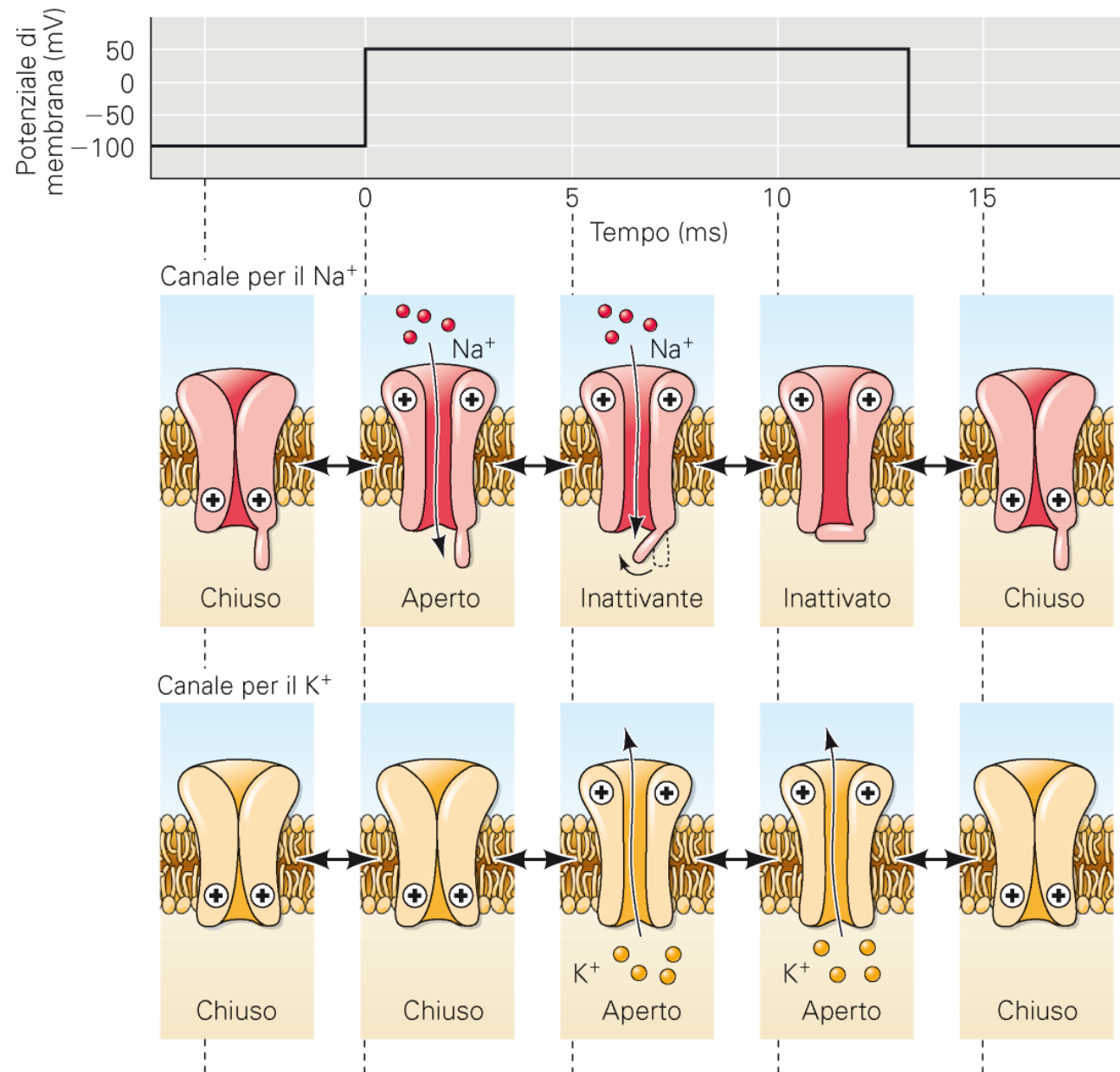


Figura 10.7 Il potenziale d'azione si genera per via dell'apertura successiva dei canali Na⁺ e K⁺ voltaggio-dipendenti. Uno dei risultati più importanti ottenuti da Hodgkin e Huxley è stata la separazione della conduttanza totale che si osserva nel corso di un potenziale d'azione, nelle sue diverse componenti ascrivibili all'apertura dei canali Na⁺ e K⁺. Il profilo del potenziale d'azione e le conduttanze che ne sono alla base possono venir calcolate dalle proprietà dei canali voltaggio-dipendenti Na⁺ e K⁺. (Modificata, con autorizzazione, da Hille 2001.)

Na⁺ Channels

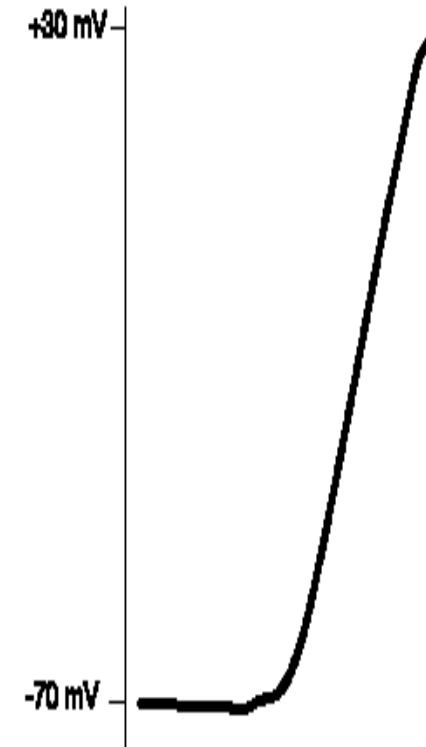
- Allow Na⁺ to enter, depolarizing the cell
- Open and close rapidly
- Have 3 conformations, and thus 3 states: closed, open, and **inactivated**

K⁺ Channels

- Allow K⁺ to exit, repolarizing the cell
- Open and close slowly
- Have 2 conformations, and thus 2 states: closed and open

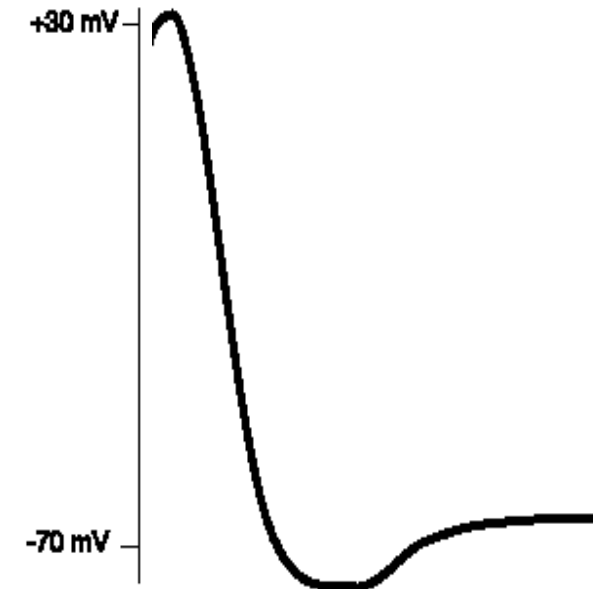
1) ASCENDING PHASE of the action potential

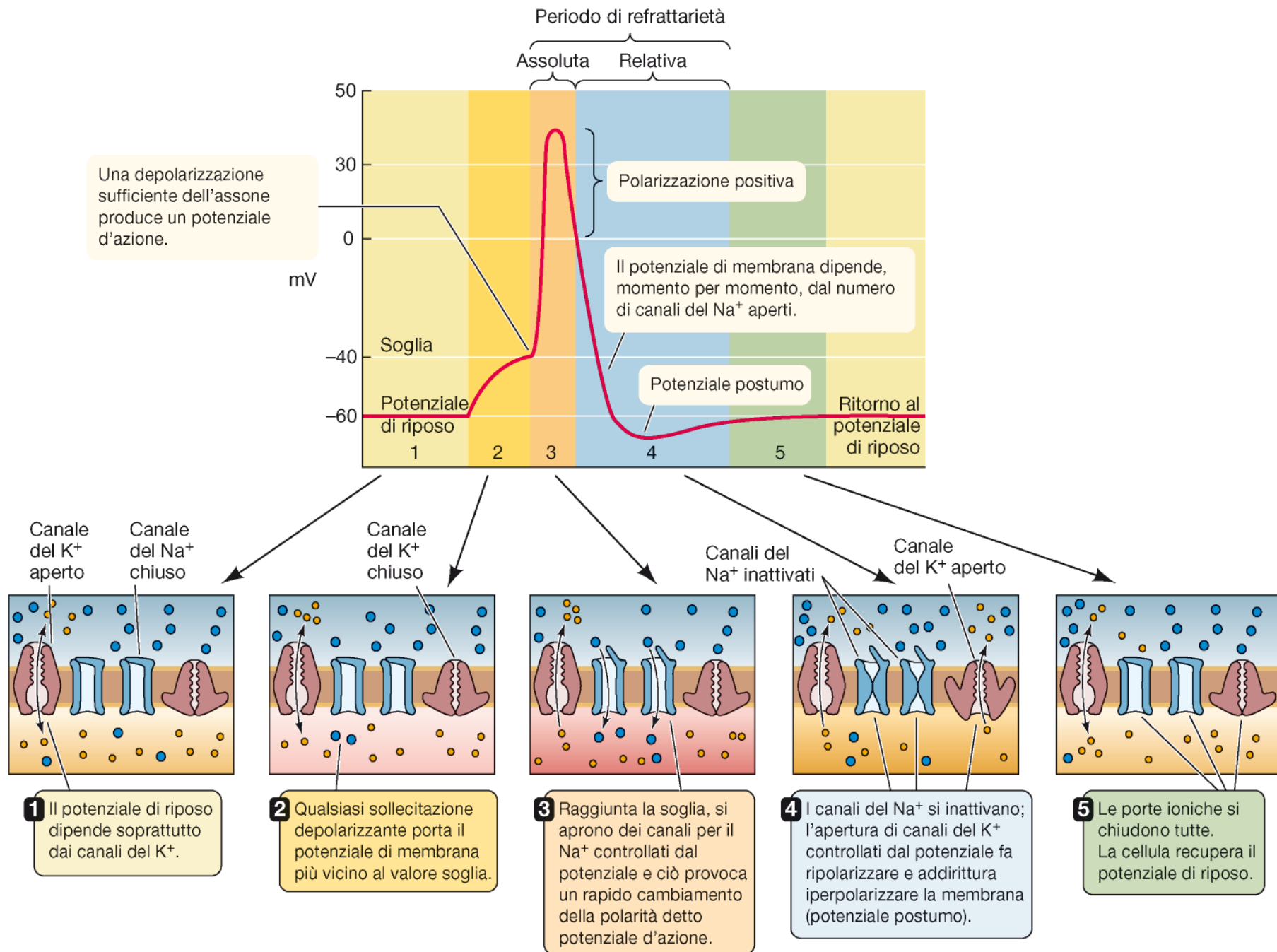
- Sodium ion influx: depolarization of the neuron
- Opening of all Na^+ channels: $E_{\{\text{Na}^+\}} = +55 \text{ mV}$ but K^+ channels also begin to open: $V_m = +30 \text{ mV}$
- Amplitude is always maximal: not variable, always up to $+30 \text{ mV}$, does not decrease with distance
- At this point, Na^+ channels begin to inactivate, ending the sodium influx into the neuron



2. FASE DISCENDENTE del potenziale d'azione

- Potassium efflux: hyperpolarization
 - Na^+ channels: inactivated
 - K^+ channels: open
- K^+ channels close slowly:
 - V_m becomes more negative
- During this phase, Na^+ channels transition from the inactivated state to the closed state.





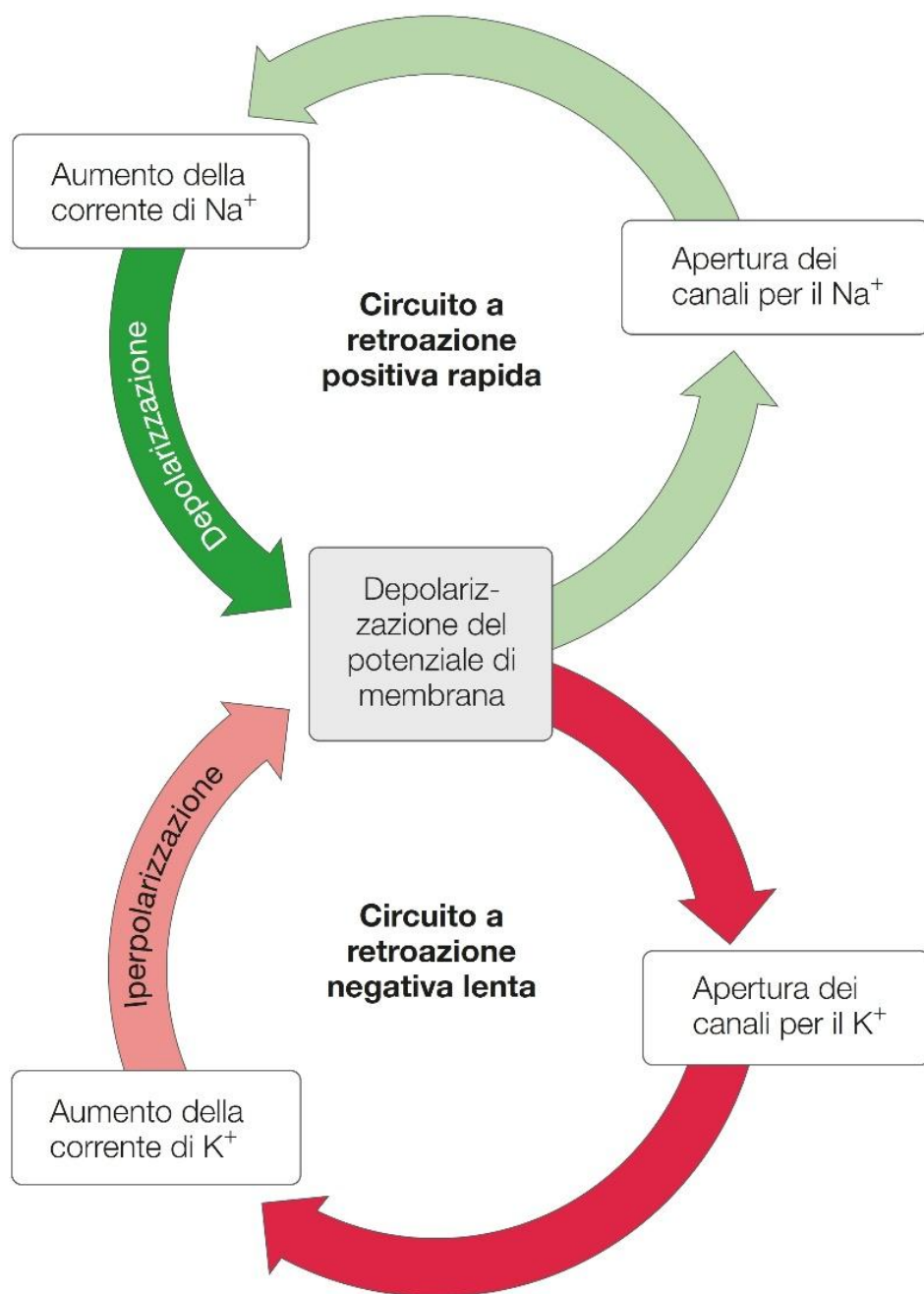
Reconstruction of the action potential (AP): REVIEW of the phases

The factor responsible for the onset of the action potential is a selective increase in Na^+ conductance.



- 1) This increase causes Na^+ influx into the neuron, depolarizing the membrane potential.
- 2) The rate of depolarization slows down when the electrochemical driving force acting on Na^+ decreases and when Na^+ conductance becomes inactivated.

- 3) At the same time, depolarization gradually activates the voltage-dependent K^+ conductance, which drives K^+ out of the cell and repolarizes the membrane potential.
- 4) K^+ conductance increases transiently, and the value of the membrane potential becomes more negative than the normal resting potential (hyperpolarization).
- 5) Hyperpolarization then causes the cessation of the voltage-dependent K^+ conductance, allowing the potential to return to its resting level.

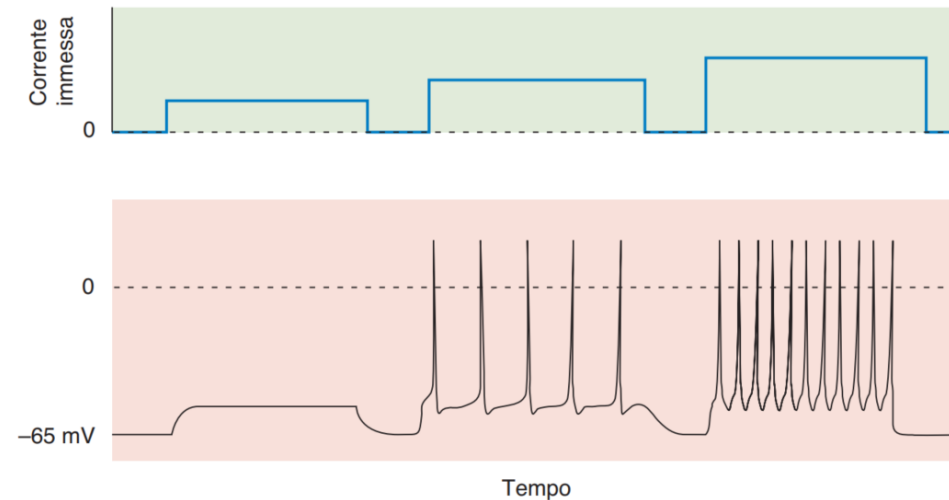
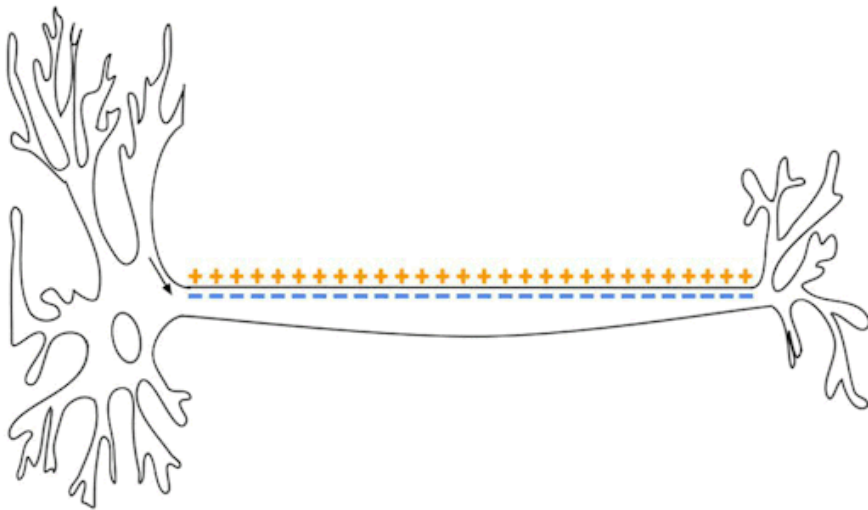


Positive feedback circuit driven by Na^+ : the action potential is self-regenerating

Negative feedback circuit driven by K^+ : returns the membrane to its resting state

Properties of the Action Potential:

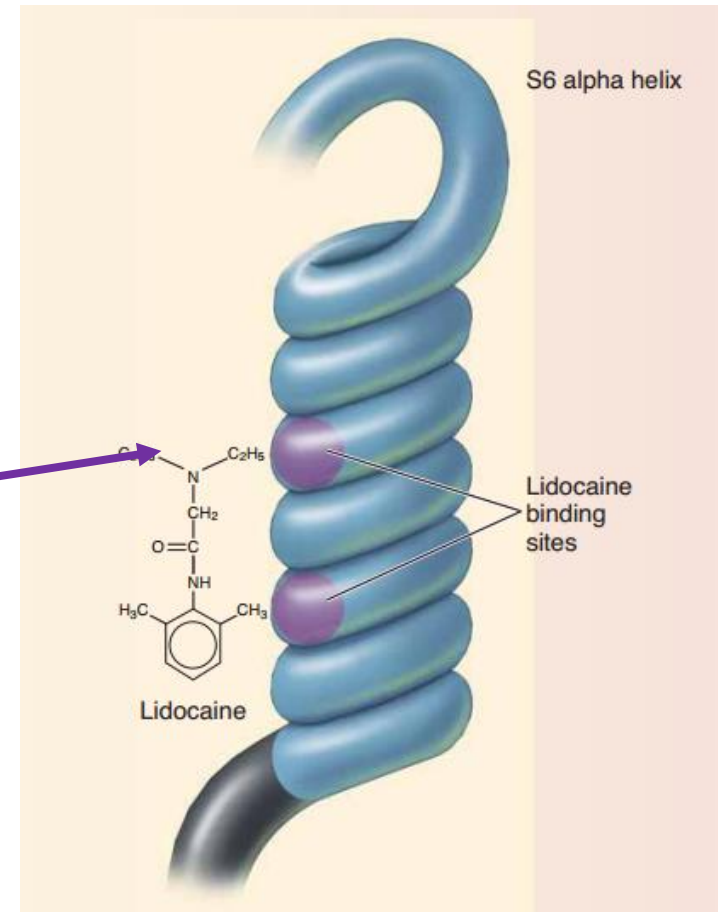
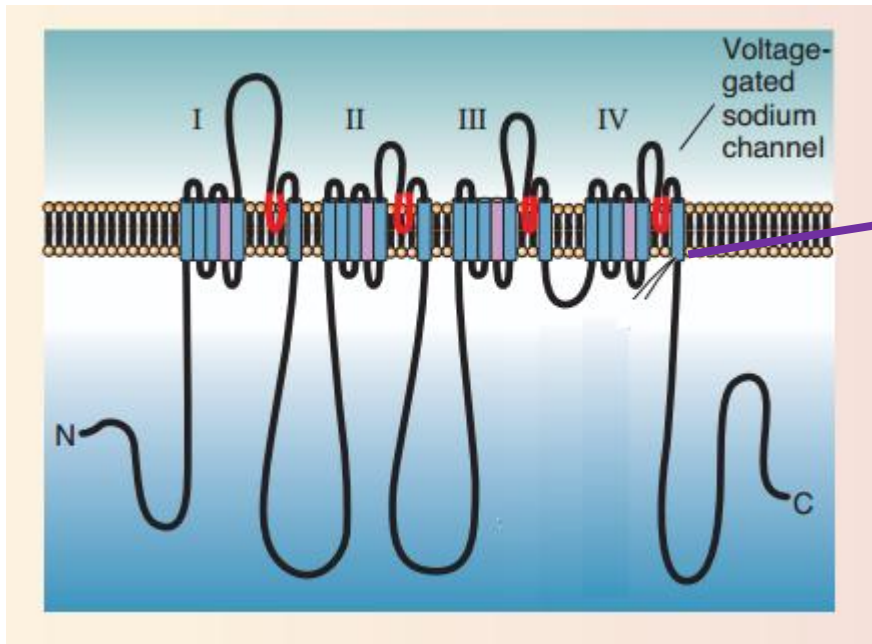
- Duration of ~ 1 ms
- The amplitude of the action potential is independent of the intensity of the current used to evoke it
- All-or-none law $\rightarrow$ action potentials occur with maximal amplitude or do not occur at all
- The intensity of a stimulus is encoded in the frequency of action potentials rather than in their amplitude



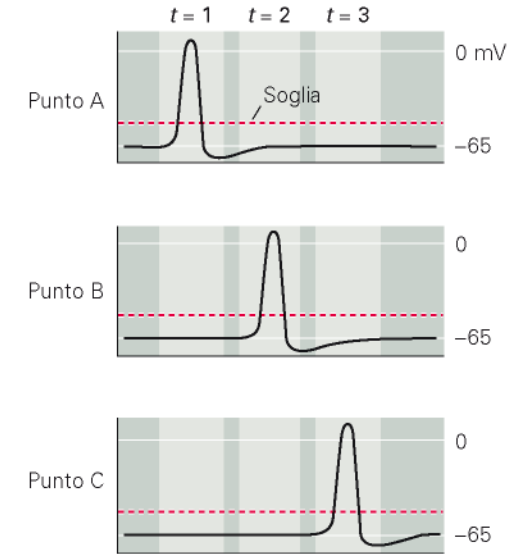
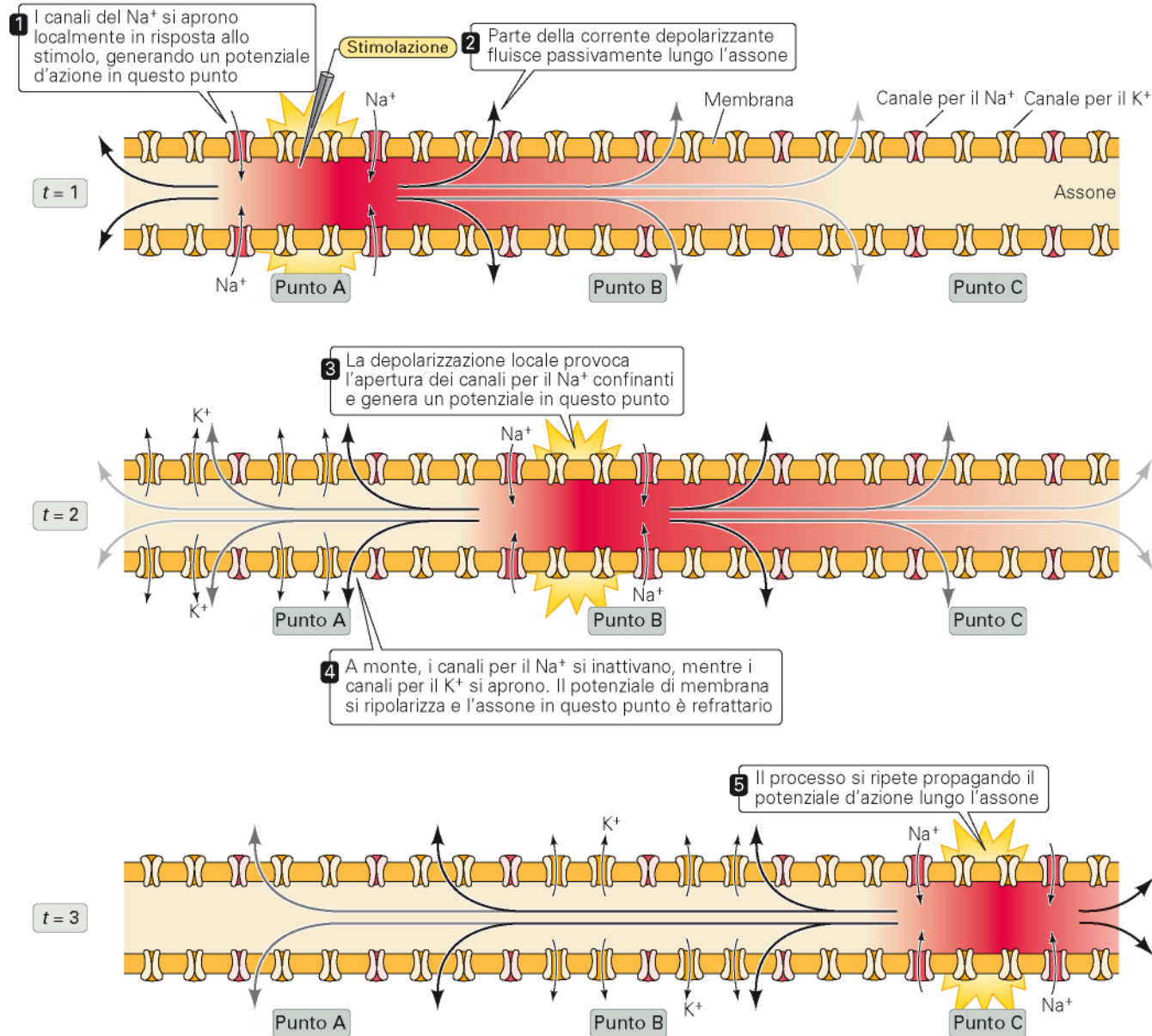
Local anesthesia

Local anesthetics reversibly block the generation and axonal conduction of the nerve impulse at the level of Na^+ channels responsible for the action potential → they prevent the changes required for Na^+ passage.

Examples: lidocaine, cocaine...



Long-distance transmission via action potentials





The propagation of action potentials requires the coordinated action of two forms of current flow:

- Passive current flow
- Active current flow (ion channels)

1. A stimulus locally depolarizes the axon, leading to the opening of voltage-dependent Na^+ channels.
2. The opening of Na^+ channels causes Na^+ influx, and the resulting depolarization of the membrane potential generates an action potential.
3. Part of the local current generated by the action potential will flow passively along the axon, in the same way that subthreshold currents propagate along the axon.
4. This passive flow of current depolarizes the membrane potential, leading to the opening of Na^+ channels.
5. The local depolarization triggers a new action potential, which then propagates in a continuous cycle until it reaches the end of the axon.

The refractory period

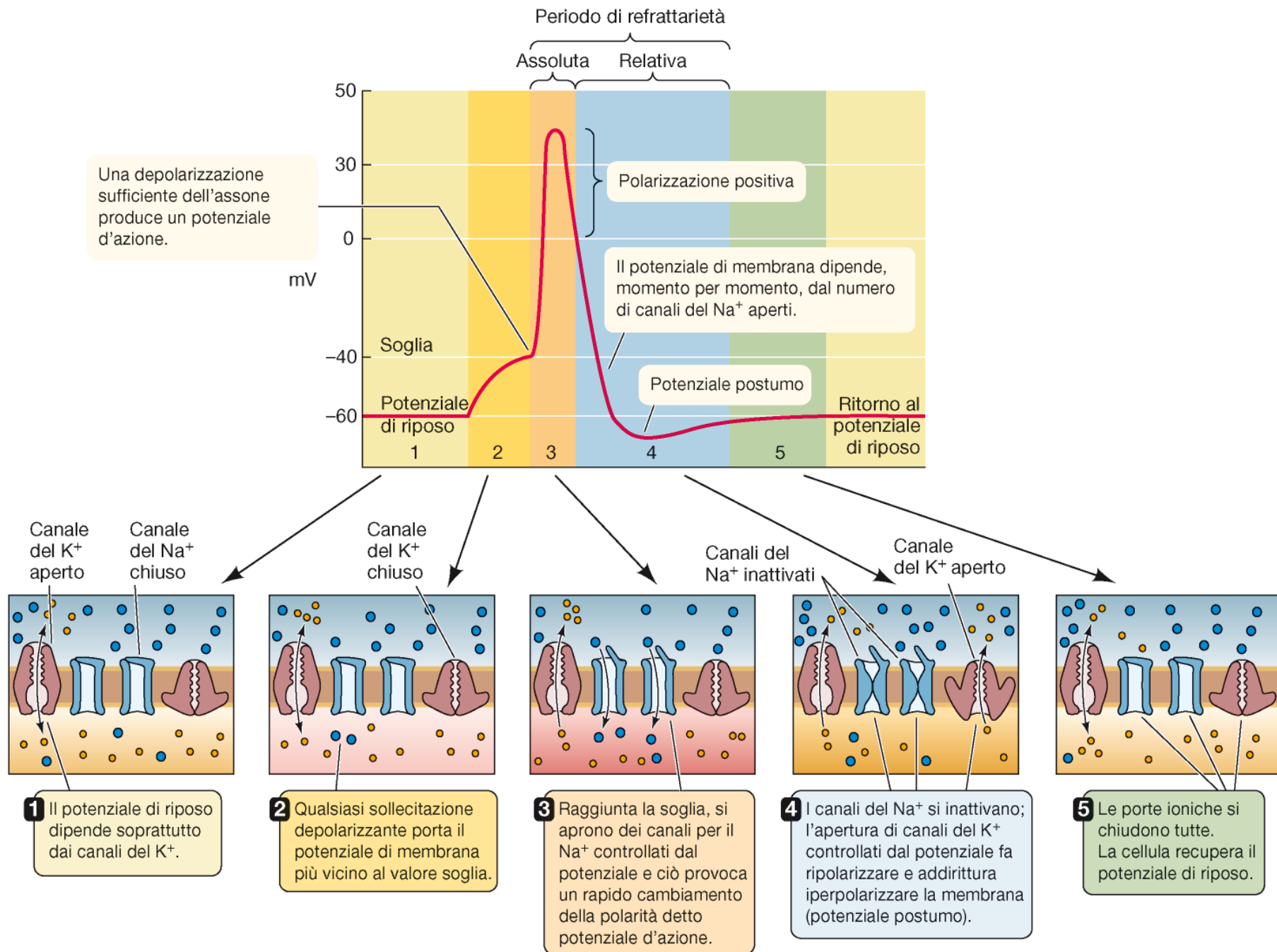
The generation of an action potential in an axon makes the generation of a subsequent potential more difficult.

This occurs because the depolarization that opens Na^+ channels also causes the delayed activation of K^+ channels and the inactivation of Na^+ channels → making it harder to produce another action potential.

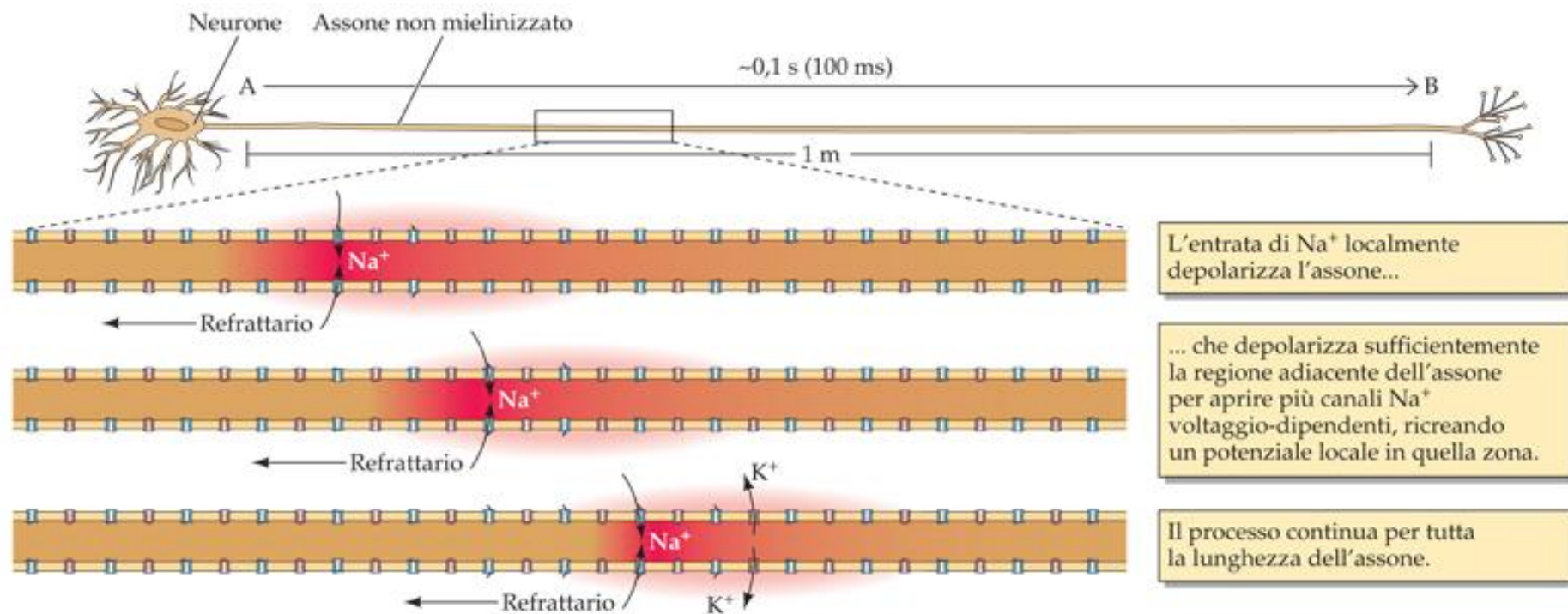
Refractoriness prevents subsequent re-excitation of the membrane, which in turn prevents action potentials from propagating backward:

- ❑ **Absolute refractory period**

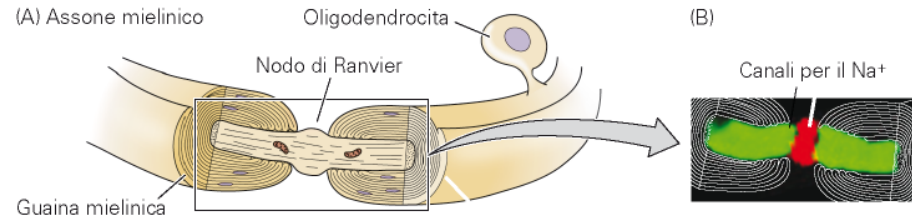
- ❑ **Relative refractory period**



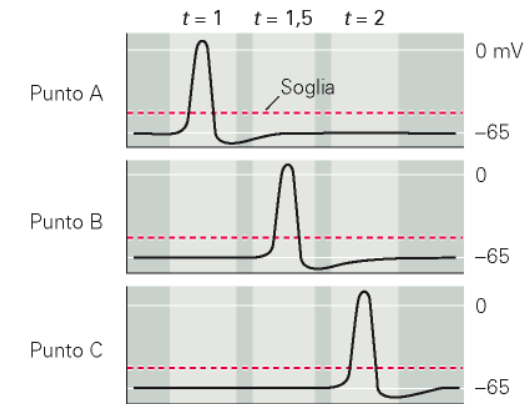
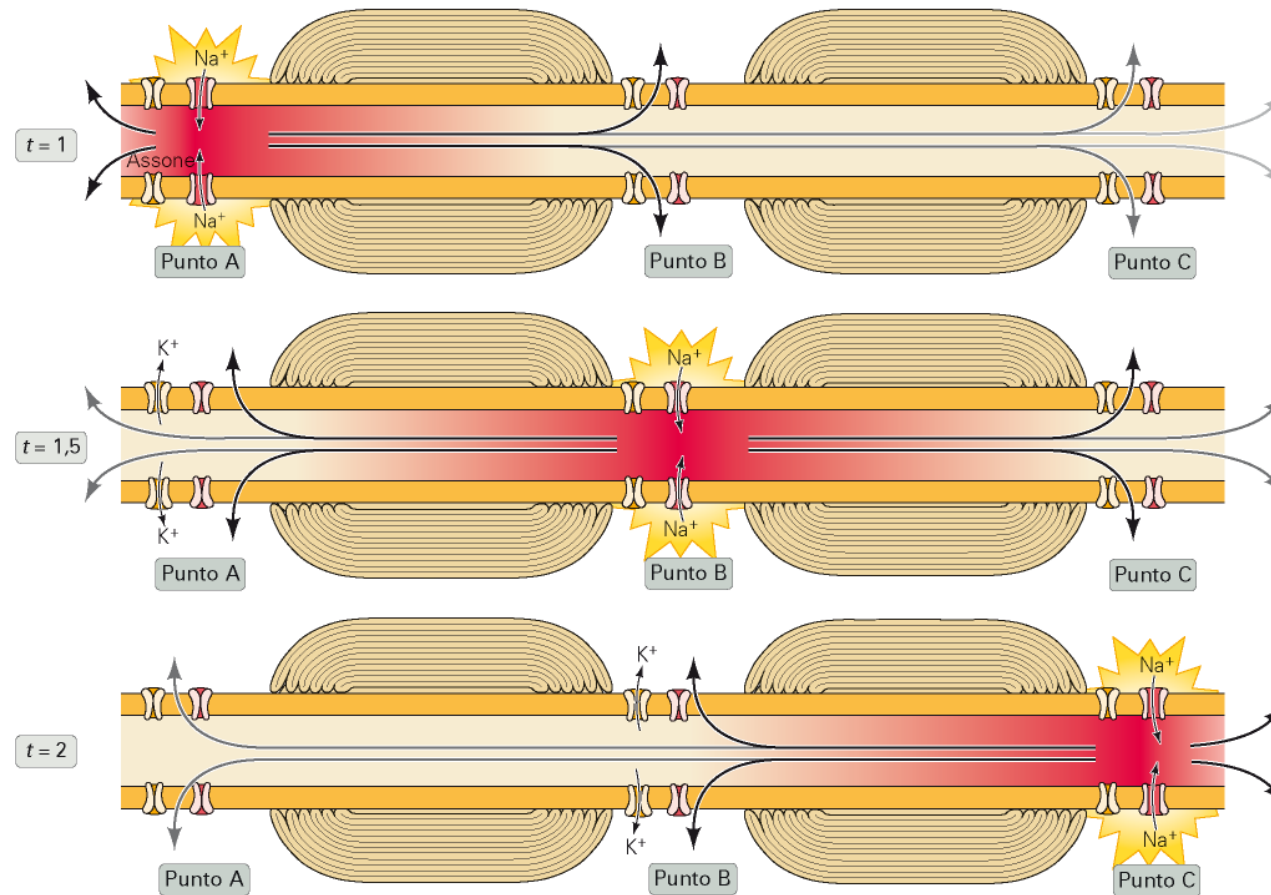
(a) Trasmissione lenta (10 m/s) del potenziale d'azione lungo un assone non mielinizzato



Propagation of action potentials in a myelinated axon

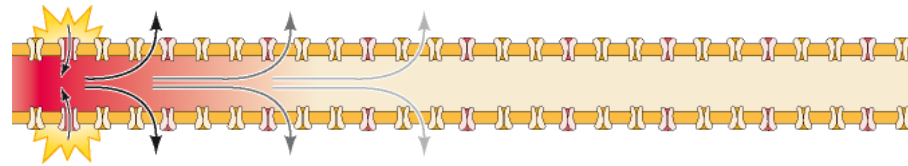


(C) Propagazione del potenziale d'azione

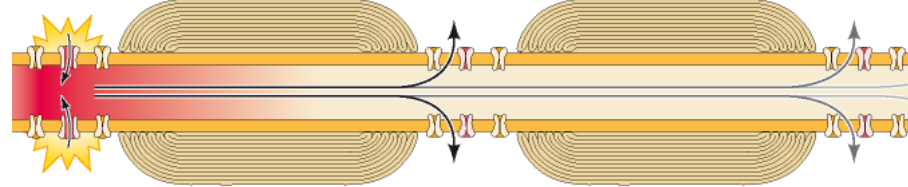


$t = 1$

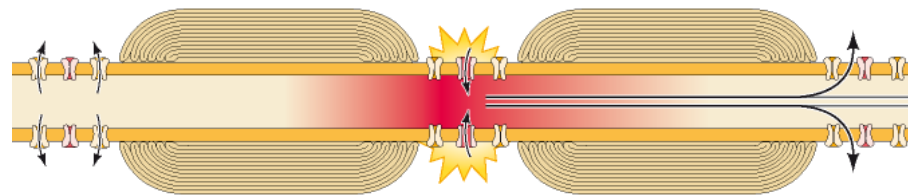
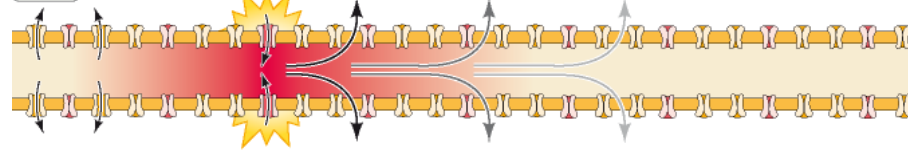
Assone amielinico



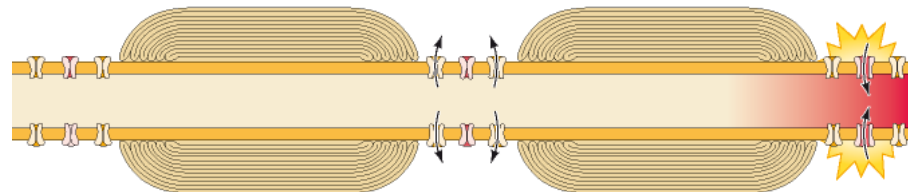
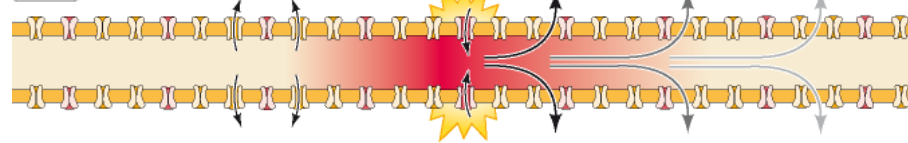
Assone mielinico



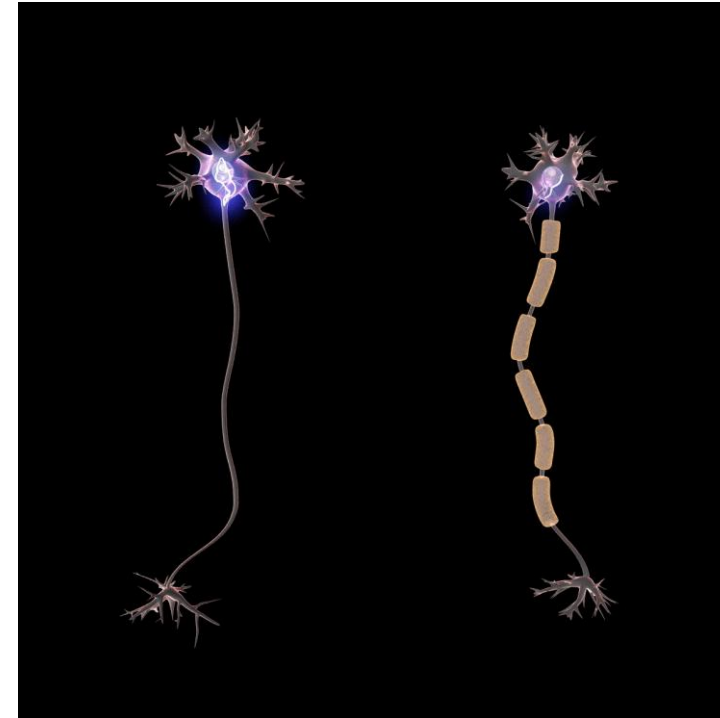
$t = 2$



$t = 3$



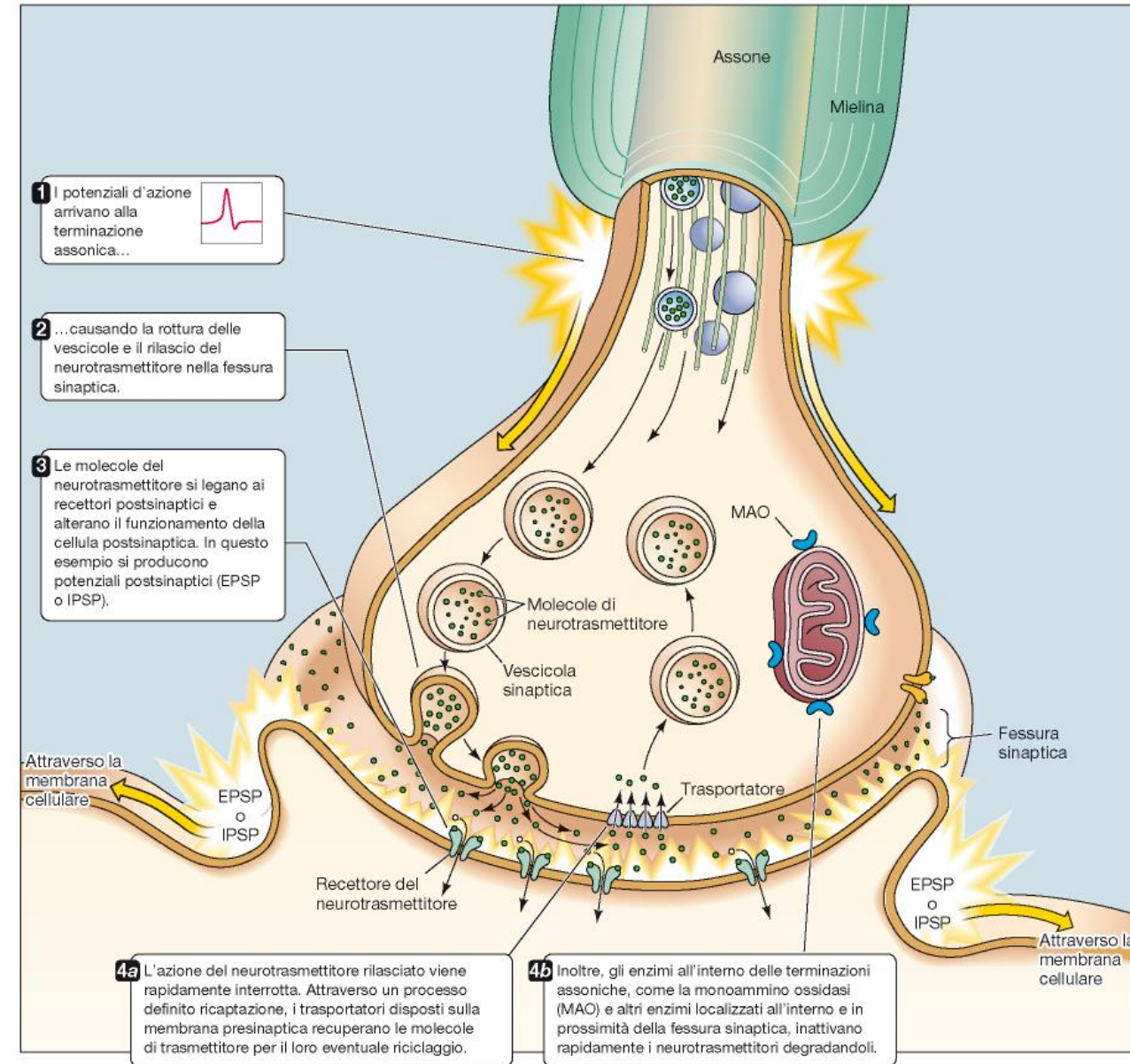
Comparison between the propagation velocities of an action potential in a myelinated and an unmyelinated axon



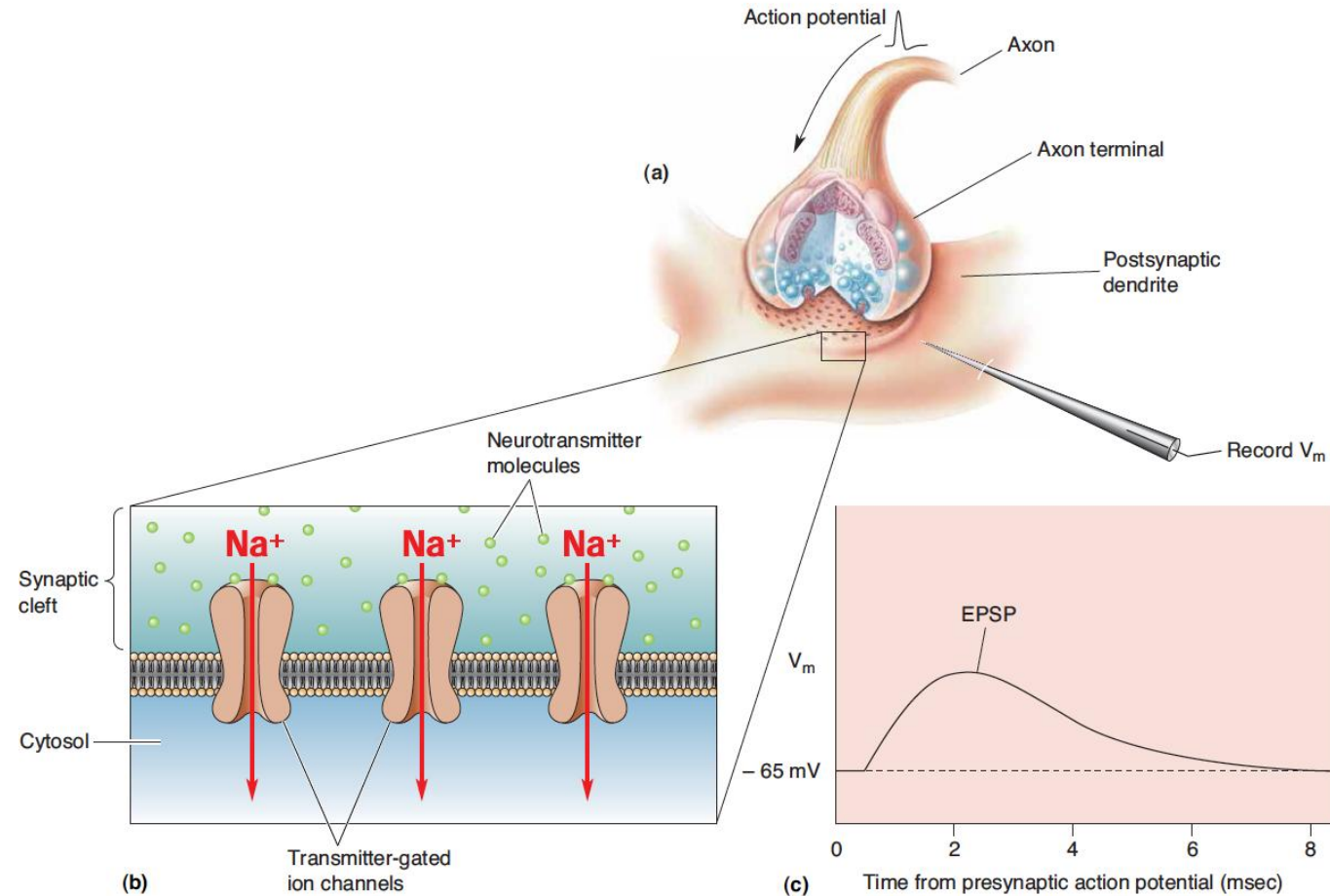
The synapse

Main events in synaptic transmission

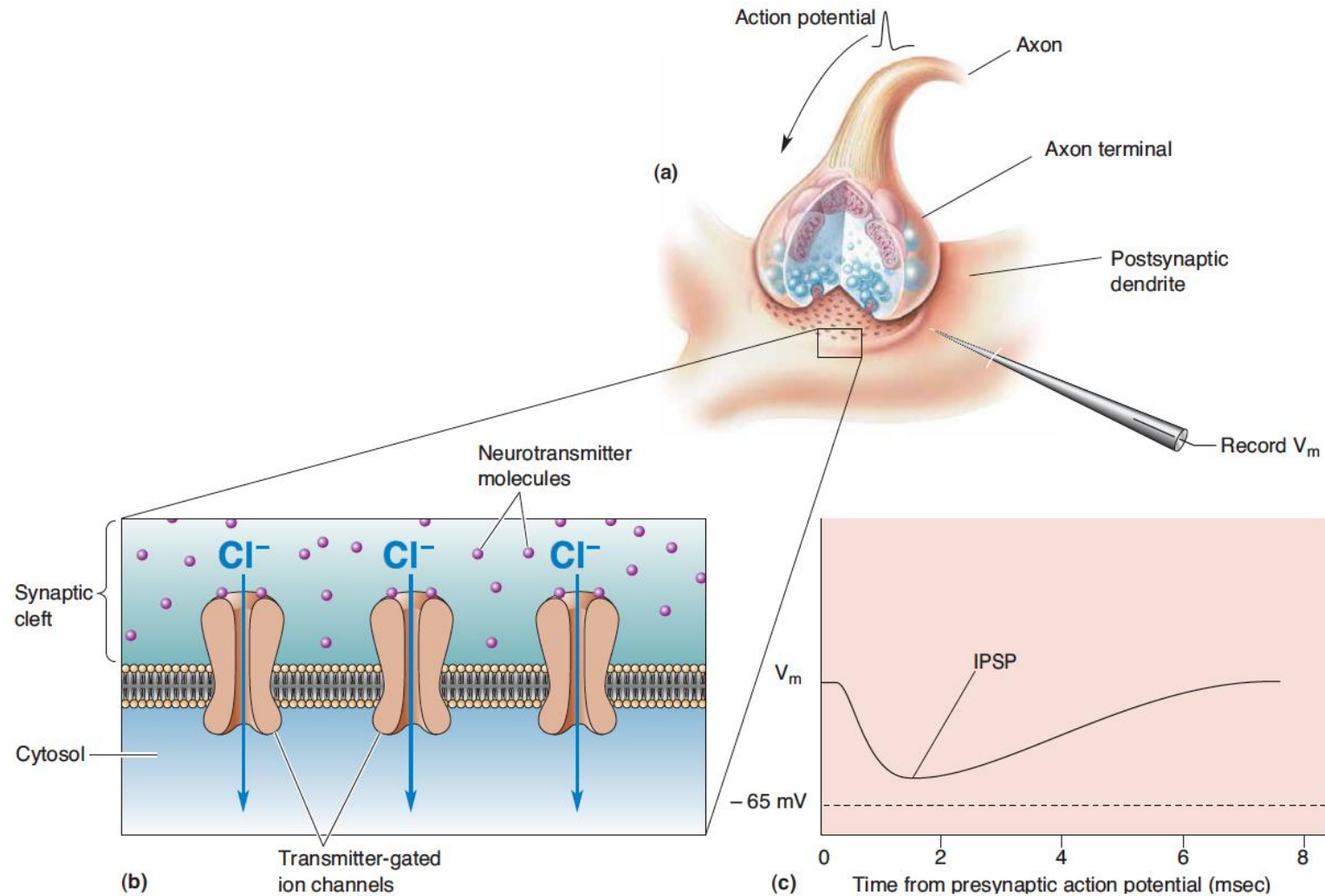
- 1. Action potential arrival** – The AP reaches the presynaptic terminal.
- 2. Calcium entry** – Voltage-gated Ca^{2+} channels open, and Ca^{2+} flows into the presynaptic terminal.
 - Vesicle fusion** – The rise in Ca^{2+} triggers synaptic vesicles to fuse with the presynaptic membrane.
- 3. Neurotransmitter release** – Neurotransmitters are released into the synaptic cleft.
- 4. Binding to receptors** – Neurotransmitters bind to receptors on the postsynaptic membrane (recaptation, degradation, ...)
- 5. Postsynaptic response** – Ion channels open or close, leading to depolarization (excitatory postsynaptic potential, EPSP) or hyperpolarization (inhibitory postsynaptic potential, IPSP).



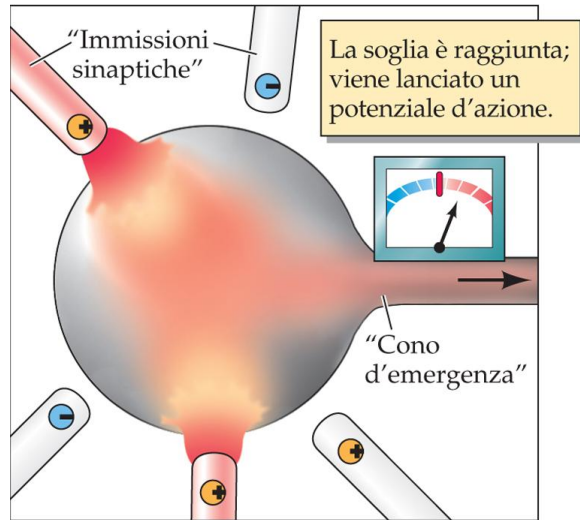
Excitatory postsynaptic potential (EPSP)



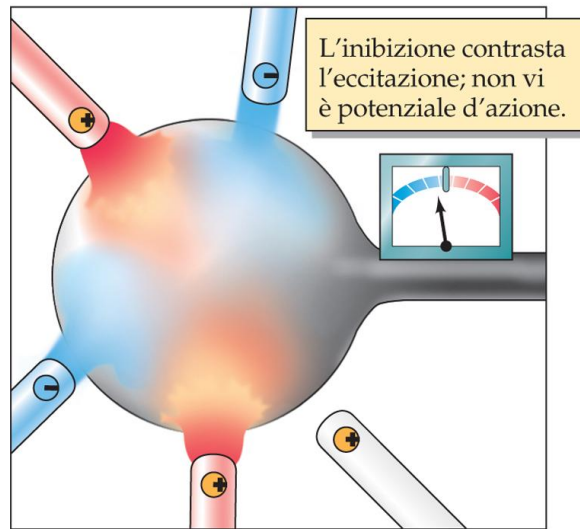
Inhibitory postsynaptic potential (IPSP)



(a) Immissioni eccitatorie inducono il potenziale d'azione nella cellula



(b) Anche l'inibizione svolge un ruolo



(c) La cellula integra eccitazione e inibizione

